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Fatty acid metabolism in T-cell function and differentiation

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REVIEW

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Abstract

Immunometabolism has recently emerged as a field of study examining the intersection between immunology and metabolism. Studies in this area have yielded new findings on the roles of a diverse range of metabolic pathways and metabolites, which have been found to control many aspects of T-cell biology, including cell differentiation, function and fate. A particularly important finding has been the discovery that to meet the energy requirements associated with their proliferation, activation and specific functions, T cells switch their metabolic signatures during differentiation. For example, whereas the induction of *de novo* fatty acid biosynthesis and fatty acid uptake programs are required for antigen-stimulation-induced proliferation and differentiation of effector T cells, fatty acid catabolism *via* β-oxidation is essential for the generation of memory T cells and the differentiation of regulatory T cells. In this review, we discuss recent advances in our understanding of the metabolism in different stages of T cells and how fatty acid metabolism in these cells controls their specific functions.

Keywords: ACC1, fatty acid biosynthesis, immunological memory

Introduction

T cells are an essential component of the adaptive immune system, acting as both modulator and effector. As such, naive T cells undergo clonal expansion and initiate effector immune responses to protect the host from pathogens. After the clearance of pathogens, a very small proportion of effector cells are maintained as memory T cells to prepare for subsequent infections (1–3). In recent years, mounting evidence has demonstrated the importance of cellular metabolism in many aspects of T-cell biology (4–7). The most prominent metabolic change following T-cell receptor (TCR)-mediated activation is the use of aerobic glycolysis, a process in which glucose is converted into lactate even though sufficient oxygen is present to support glucose catabolism *via* the tricarboxylic acid (TCA) cycle and oxidative phosphorylation (OXPHOS) (8). To achieve this metabolic change, T-cell co-stimulation *via* CD28 and TCR engagement activates two signaling pathways: the PI3K–Akt (phosphatidylinositol 3-kinase–protein kinase B) signaling pathway and the mammalian target of rapamycin (mTOR) signaling pathway. In activating T cells, the PI3K–Akt signaling axis is required for the induction of Glut1 expression and subsequent glycolysis (9). Downstream of the PI3K-Akt signaling pathway, mTOR kinase integrates signals associated with nutrient levels, energy status and cell stress responses, and it can induce multiple outcomes in cells including cell growth, proliferation and changes in metabolic programs (8, 10). Thus, the PI3K–Akt–mTOR signaling axis is understood to be the primary regulator of early metabolic reprogramming associated with T-cell activation and differentiation (11, 12).

To meet the energy requirements associated with their activation and rapid proliferation, T cells switch their metabolic program from fatty acid (FA) β-oxidation (FAO) and catabolic metabolism to aerobic glycolysis and anabolic metabolism, respectively (13). Beyond their role in energy production in the form of ATP, the synthesis and accumulation of metabolic intermediates are prerequisite for changes in biomass and cell size during T-cell differentiation (7, 13). Increased rates of glucose and amino acid uptake lead to the generation of metabolic intermediates that modulate the activity of several key signaling effectors. For example, levels of glycolytic intermediates alter the activity of RNA-binding proteins, regulate post-translational glycosylation and activate the key metabolic regulator AMPK (AMP-activated protein kinase) (14). Amino acid metabolism acts as a potent activator of mTORC1

(mTOR complex 1) signaling (15). Lipid species control the activity of several key signaling molecules of T-cell activation. Many core metabolic pathways including glycolysis, FAO and FA biosynthesis are highly affected during T-cell differentiation, and a series of recent studies has revealed that FA metabolism is a crucial component of T-cell differentiation (4, 11, 16–18).

Here, we review recent discoveries highlighting the impact of FA metabolism on T-cell differentiation and function. We first summarize the different metabolic profiles of effector T-cell subsets and memory T cells and then describe in detail how metabolic reprogramming, specifically cellular FA metabolism, controls the transition of activated effector cells and quiescent memory T cells. We then provide new insights into how these FA metabolic programs interplay with immunological signals. Finally, we conclude with a discussion of future directions and the therapeutic implications of modulating FA metabolism in T cells.

Immunometabolism in T-cell activation

Upon activation, T cells reprogram their metabolic pathways to meet the bioenergetic demand for rapid proliferation. The process towards an anabolic phenotype during T-cell activation is regulated by several signaling pathways and transcription factors. T-cell stimulation *via* CD28 co-stimulation with TCR engagement drives rapid proliferation through activation of the PI3K–Akt and mTOR pathways (11, 12). Within the first 24–30 h after anti-TCR/CD28-mediated stimulation, T cells generate markedly increased amounts of the metabolites involved in the anabolic pathways of amino acids, polyamines, carbohydrates, nucleotides and lipids (8, 19). TCR stimulation–dependent metabolic reprogramming is accomplished *via* dynamic changes in the expression of metabolic enzymes downstream of mTOR activation and the induction of the transcription factor c-Myc (8, 19). c-Myc is critical for the induction of the metabolic reprogramming that occurs early in the timeline of activated T cells. Indeed, c-Myc–deficient activated T cells show a marked reduction in proliferative capacity and cell growth, in part due to impaired glucose and glutamine metabolism (8, 19). mTOR also controls cellular metabolism *via* modulation of hypoxia-inducible factor 1 subunit alpha (HIF1a) in some contexts (20). Genetic deletion of Hif1a in T-helper 17 (Th17) cells or activated $CD8^+$ T cells, but not other T-cell subsets, leads to a reduction of the expression of *Glut1* and other metabolic genes involved in glycolysis (20, 21). These observations imply that Myc-dependent or HIF1a-dependent metabolic reprogramming plays a pivotal role in activation-induced T-cell proliferation and growth (8, 19).

In addition to glucose metabolism, other metabolic pathways such as lipid and mitochondrial metabolism are now also recognized as being critical aspects of T-cell activation. For example, a series of recent studies has demonstrated the crucial requirement for FA metabolism in T-cell differentiation (4, 18). During T-cell differentiation, the mTOR pathway functions primarily through the post-transcriptional regulation of key metabolic transcription factors. Indeed, T cells deficient in Raptor, an essential component of mTORC1, exhibit impaired lipid biosynthesis accompanied by diminished protein levels of two primary regulators of lipid biosynthesis: sterol regulatory element-binding protein 1 (SREBP1) and SREBP2 (18, 22, 23). SREBPs play an essential role in metabolic reprogramming in response to mitogenic stimulation in $CD8^+$ T cells (18). Kidani *et al.* have reported that inactivation of SREBPs *via* genetic deletion of SREBP cleavage-activating protein (SCAP) results in diminished T-cell proliferative capacity both *in vitro* and *in vivo* during viral infection (18). They also demonstrated that functional loss of SREBPs not only attenuates lipid synthesis but also reduces the prevalence of metabolic reprogramming towards a glycolytic phenotype in activated $CD8^+$ T cells. Interestingly, supplementation of *Scap*-deficient $CD8^+$ T cells with cholesterol restored their growth and proliferative capacity, suggesting that intracellular cholesterol serves as a metabolic checkpoint during activation-induced blastogenesis (18). These data are consistent with another finding indicating the crucial role of liver X receptors in the distribution and availability of intracellular cholesterol for cell growth and proliferation in activated T cells (24). Notably, SREBP activity is not required under homeostatic proliferation of naive or memory $CD8^+$ T cells (18). Together, these findings suggest that lipid biosynthesis pathways modulate mTOR-mediated metabolic reprogramming during T-cell activation.

In addition to FA biosynthesis, we have reported that activated $CD4^+$ T cells utilize exogenous FAs as nutrients via up-regulation of peroxisome proliferator activated receptor gamma (PPARγ) and its target genes that are associated with the uptake of FAs (11). Unsaturated FAs and other FA metabolites such as 15-deoxy-Δ12,14-prostaglandin J2 are suspected ligands for PPARγ in some cell types including adipocytes (25). Therefore, these FA metabolites may bind to PPARγ and control its target genes involved in FA uptake in activated $CD4^+$ T cells. Thus, both FA uptake and biosynthesis pathways are required for the early and full activation of naive or memory $CD4^+$ T cells in response to TCR stimulation in murine and human systems.

FA metabolic control of Th17 and regulatory T-cell differentiation

Maintenance of the balance between protective immunity and chronic inflammation requires that T cells appropriately differentiate into the effector and regulatory cell lineages. Th17 cells and regulatory T (Treg) cells represent two distinct phenotypes of $CD4^+$ T cells with completely different functions. Whereas Th17 cells have a pro-inflammatory phenotype, Treg cells have an anti-inflammatory one. Thus, the Th17/Treg balance is crucial for preventing excessive immune responses, autoimmune disorders and metabolic syndrome pathogenesis (26, 27).

Th-cell differentiation from naive T cells is controlled by the cytokine microenvironment. As with T-cell activation, metabolic control is also necessary for Th subset differentiation (28). It is largely accepted that Th1, Th2 and Th17 effector cells are highly dependent on glycolysis, whereas Treg cells preferentially use FAO (13). In addition, the specific function of Th1 cells is highly dependent on environmental levels of FA nutrients. For example, long-chain FAs, including capric acid and lauric acid, enhance Th1 cell differentiation *via* the p38-MAPK pathway, exacerbating the disease state

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in mice with experimental autoimmune encephalomyelitis (EAE) (29). In contrast, polyunsaturated FAs (PUFAs) such as docosahexaenoic acid down-regulate cytokine expression associated with Th1 cells (30).

As already noted, Treg cells predominantly use mitochondrial FAO for their development and survival, whereas the differentiation of Th17 cells is heavily dependent on *de novo* FA biosynthesis (17, 31). Th17 cells cultured with IL-23 and IL-1β show a more inflammatory phenotype and this results in up-regulation of genes involved in FA biosynthesis (32). Furthermore, the TGFβ–Smad2/3 pathway induces expression of fatty acyl-CoA synthetases such as *Acsl6* and *Acsbg1* in Treg cells (33). Inhibition of FA biosynthesis, either by the pharmacological inhibitor 5-tetradecyloxy-2-furoic acid (TOFA) or by causing deficiency of acetyl-CoA carboxylase 1 (ACC1), specifically in CD4⁺ T cells, significantly inhibits Th17 cell differentiation (17). Similarly, the proportion and number of IL-17A–producing memory phenotype CD4⁺ T cells have also been shown to be decreased in mice deficient in ACC1 (*Acaca*$^{\Delta T}$ mice). ACC1, the gene product of *Acaca*, is a key biotinylated enzyme that catalyzes the ATP-dependent carboxylation of Ac-CoA (acetyl-CoA) to malonyl-CoA and regulates cellular FA metabolism (34). The deletion of ACC1 in mice results in embryonic lethality, indicating that FA synthesis is essential for embryonic development (35). In activated CD8⁺ T cells or many tumor cell types, the mTOR–SREBP pathway is activated to induce expression of genes, such as *Acaca*, encoding molecules related to FA biosynthesis (18, 23).

Interestingly, FA fluxes have also emerged as an important regulator of Th17 plasticity. Pharmacological inhibition or genetic deletion of ACC1 increases expression of the transcription factor FOXP3, which is a marker of Treg cells, in CD4⁺ T cells cultured under Th17-skewing conditions (17, 31, 36). This shift towards Treg cell differentiation in ACC1-deficient cells can be overcome by extrinsic supplementation with exogenous FAs such as palmitate or oleate, suggesting that FAs or FA products play an essential role in regulating the Th17/Treg balance (Fig. 1, left). Consistent with these *in vitro* results, administration of TOFA or genetic deletion of ACC1 protected mice from developing the EAE pathology and reduced the proportion of IL-17A-producing CD4⁺ T cells in the central nervous system (17, 31). Furthermore, TOFA administration ameliorated the EAE pathology even in mice with diet-induced obesity.

Recent RNA-sequencing studies performed using T cells from EAE mice have demonstrated that low levels of CD5 molecule-like (CD5L) are correlated with Th17 pathogenicity (37, 38). CD5L has been shown to regulate FA flux through the *de novo* FA biosynthetic pathway (37, 38). Lipidomics analyses in CD5L-deficient Th17 cells revealed global differences in the FA composition of the lipids both in phospholipids and neutral lipids. Analyses of CD5L-deficient Th17 cells revealed increased levels of saturated FAs and monounsaturated FAs

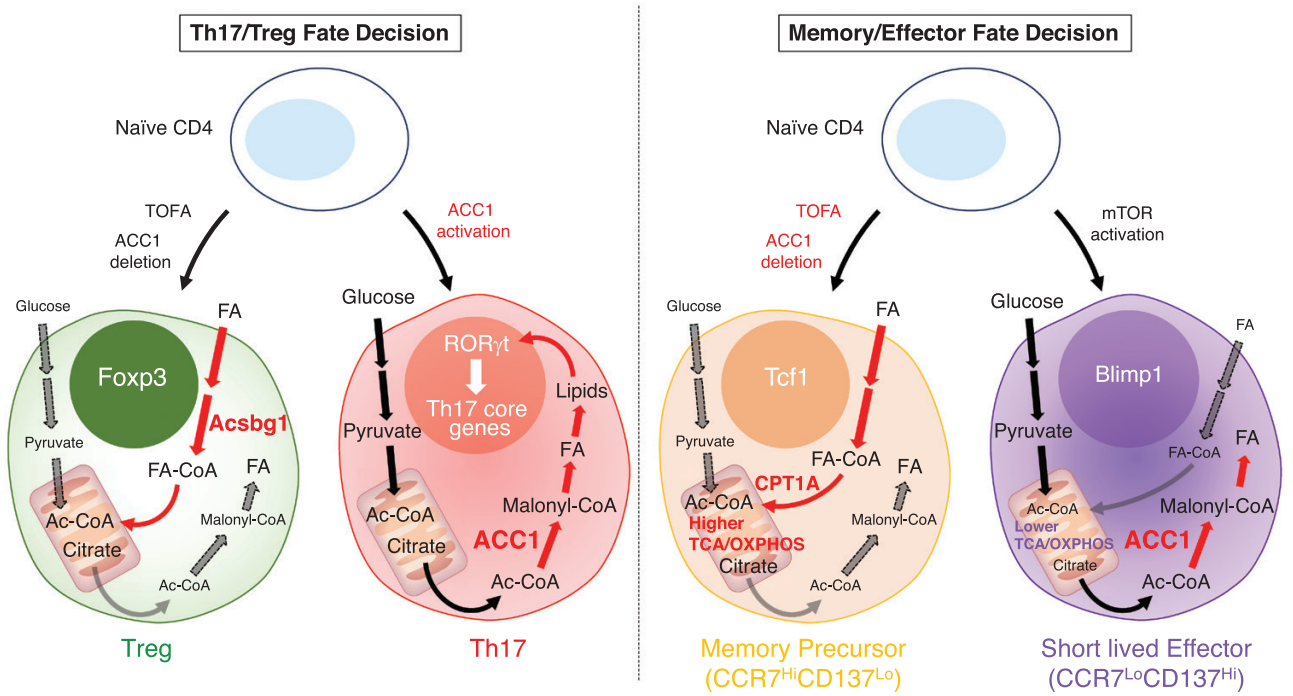

Fig. 1. Metabolic control of Th17/Treg and memory/effector cell fate decisions. (Left) Th17 cell differentiation is heavily dependent on *de novo* FA biosynthesis. Inhibition of FA biosynthesis *via* ACC1 deficiency in CD4⁺ T cells inhibits Th17 cell differentiation. Pharmacological inhibition or genetic deletion of ACC1 increases FOXP3 expression in CD4⁺ T cells cultured under Th17-skewing conditions. This shift towards Treg cell differentiation in ACC1-deficient cells can be overcome by supplementation with exogenous FAs. Thus, FA metabolism plays an essential role in regulating the Th17/Treg balance. (Right) Development towards a memory T-cell phenotype is marked by a metabolic switch from active metabolism to catabolism of FAs via FAO. Genetic deletion or pharmacological inhibition of ACC1 during the effector phase results in increased memory CD4⁺ T-cell formation via alteration of the intrinsic metabolic profiles of effector T cells to resemble that of memory T cells. Therefore, FA biosynthesis controls the CD4⁺ T-cell effector/memory fate decision during early differentiation from naive to effector T-helper cells.

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(MUFAs) and a corresponding decrease in PUFAs, such as C20:4 (38). Furthermore, exogenous addition of PUFA to Th17 cells decreased binding of retinoic acid-related orphan receptor γt (RORγt, which is crucial for the development of T cells, especially Th17 cells) at *Il17* and *Il23r* loci, whereas providing exogenous saturated FAs or MUFAs increased RORγt binding to the target gene loci (38). Thus, the intracellular FA composition appears to impact Th17 function, at least in part, *via* RORγt recruitment and/or activity at the signature cytokine loci.

Other instances of cellular metabolism that affect Th17 cell differentiation can be found in cholesterol metabolism, indicating the importance of fluxes in cholesterol biosynthesis in the control of RORγt activity. Genetic deletion of specific enzymes in the distal branch of the cholesterol biosynthetic pathway reduces IL-17 production in $CD4^+$ T cells (39, 40). In addition, treatment of Th17 cells with statins, which are pharmacological inhibitors of 3-hydroxy-3-methylglutaryl CoA reductase (which is part of the cholesterol biosynthetic pathway), also affects the Th17/Treg balance (41). Although oxysterols and cholesterol biosynthetic intermediates, including lanosterol, have the ability to increase RORγt activity, it remains unclear which specific sterol metabolites serve as the physiologic ligand(s) of RORγt during Th17 cell differentiation. Interestingly, neither pharmacological inhibition nor genetic deletion of cholesterol biosynthesis was able to completely ablate IL-17 production (39, 40). Although it is not entirely clear which molecular mechanisms link lipid homeostasis and RORγt function, this previous finding and those discussed above regarding FA balance controlling RORγt transactivation activity suggest that FA and cholesterol metabolites function together as physiologic ligands of RORγt.

The metabolic processes of Treg cells are markedly different from those of Th17 cells. As already discussed, recent studies have demonstrated that mitochondrial FAO plays an essential role in the suppressive function of Treg cells in the tumor microenvironment (42). Consistently, another recent study has also shown that Tregs rely on FAO and OXPHOS for their homeostasis and suppressive capacity (43). FAO involves the degradation of FAs by sequential removal of two-carbon units from the acyl chain to produce Ac-CoA, which then enters the mitochondrial TCA cycle to regulate OXPHOS (Fig. 1, left) (33, 44). Our recent data show that lung Treg cells effectively acquire exogenous FAs to fuel OXPHOS, and that Acsbg1-mediated acyl-CoA synthesis serves as a novel metabolic checkpoint controlling Treg cell homeostasis (33). Indeed, the genetic deletion of Acsbg1 caused mitochondrial dysfunction in Treg cells. Carnitine palmitoyltransferase 1 (CPT1), present in the outer mitochondrial membrane, catalyzes the esterification of long-chain FAs with carnitine to form acyl-carnitine (44). The rate of FAO depends greatly on the entry of acyl groups into the mitochondria, and thus depends mainly on the activity of CPT1. Consistent with the finding above, Treg development was abrogated by treatment with etomoxir, an inhibitor of CPT1, suggesting that Treg cells depend on this pathway for their development (45).

FA metabolism in the generation of memory T cells

Immunological memory is a hallmark of adaptive immunity, with the quality of the adaptive immune response highly dependent on the size of the antigen-specific memory T-cell pool. Clarifying the molecular events governing memory T-cell generation may help the development of improved strategies for the design of vaccines and immunotherapeutics (3, 5). Metabolic profiling has revealed that, similarly to naive T cells, memory T cells use FAO and OXPHOS as the primary processes through which they meet their energetic needs (7, 46). Molecularly, the development towards a memory T-cell phenotype is marked by a switch from active metabolism to the catabolism of FAs via FAO. This metabolic switch from glycolysis to FAO is essential for the development and survival of memory $CD8^+$ T cells (13, 46). In fact, blockade of glycolysis supports memory $CD8^+$ T-cell formation, suggesting that graded levels of glycolytic flux in the effector phase can act as a metabolic rheostat to control $CD8^+$ T-cell fate (47).

Inhibition of mTOR, the master kinase that regulates cellular metabolism via glycolysis, glutaminolysis and FA biosynthesis, enhances the generation of memory $CD8^+$ T cells (48). Inhibition of mTOR and activation of AMPK (which inhibits FA biosynthesis and activates lipid catabolism) also strongly stimulate the catabolic process of autophagy, which is essential for T-cell memory and persistence; indeed, genetic deletion of Atg5 or Atg7 (which are essential for autophagy) results in impaired memory T-cell formation (49, 50). In a different study, the pharmacological activation of AMPK signaling using metformin was demonstrated to promote T-cell persistence *in vivo* (51, 52). Signaling pathways that drive catabolic programs, such as those stimulated by IL-15 and IL-7, support mitochondrial metabolism, and CPT1a expression and FAO have been found to promote T-cell longevity *in vivo* (53). As a result, memory T cells have increased mitochondrial mass and greater spare respiratory capacity compared with those of naive or effector T cells, which confers them with an energetic advantage for survival.

As already discussed, the changing metabolic demands as the cell transitions from the effector phase to the memory phase play a pivotal role in the formation of memory T cells. Upon activation via antigenic stimulation and co-stimulatory signals, naive T cells undergo clonal expansion and differentiate into effector populations. Conceptually, effector $CD8^+$ T cells can be divided into at least two distinct subpopulations: short-lived effector cells (SLECs) and memory precursor effector cells (MPECs) (54–56). MPECs are identified as the IL-$7Ra^{hi}KLRG1^{lo}$ phenotype and are able to survive to become long-lived memory T cells. In contrast, SLECs are IL-$7Ra^{lo}KLRG1^{hi}$ and easily succumb to apoptosis following clearance of an infection. The SLEC/MPEC balance can be modulated by altering the duration of cognate antigen stimulation, the composition of the cytokine milieu or by co-stimulatory signals (54–56).

The cytokine milieu plays an essential role in T-cell activation. The duration and strength of IL-2 signals control the proliferative capacity of T cells by modulating T-cell metabolism, with IL-2 stimulating expression of the transcription factor c-Myc and activation of mTORC1 (57, 58). It has also been suggested that $CD8^+$ T cells undergo asymmetric division to give rise to daughter cells that are differentially fated toward the effector and memory lineages (59, 60). Asymmetric cell division and asymmetric activation of mTOR, Myc and PI3K at the early phase of cell division can affect cell fate

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determination of the SLEC/MPEC phenotype. For example, the distal daughter cells to antigen-presenting cells have an increased tendency to acquire a memory phenotype, whereas the daughter cells proximal to the antigen-presenting cells are more likely to acquire the SLEC phenotype (58–62). Furthermore, effector CD8$^+$ T cells with a reduced mitochondrial membrane potential in addition to reduced levels of Myc-mediated glycolysis and mTOR activation have great potential to differentiate into memory T cells (61).

Relatively little is known about the influence of cellular metabolism on memory CD4$^+$ T-cell formation compared with that of memory CD8$^+$ T cells, but recent single-cell analyses have shown that effector CD4$^+$ T-helper cells can be divided into two distinct populations based on the expression of the FA biosynthetic genes, including those of Acaca, Acsl3, Fads2 and Mcat (16). Interestingly, in contrast to the FA biosynthesis genes, those associated with glycolysis appear to show more homogeneous expression profiles in effector T-helper cells. Single-cell analyses of effector T-helper cells have shown that Acaca expression is negatively correlated with the memory signature genes Klf2, Ccr7, Foxo1 and Tcf7, whereas it is positively correlated with those of Ltb4r1, Tnfrsf9 and Casp9 (16).

Klf2, Ccr7 and Tcf7 are associated with the memory T-cell population, and Casp9 and FA biosynthetic enzymes are strongly induced in effector cells (16). A previous study has shown that anti-viral Th1 cells could be distinguished based on expression of Ly6C and T-bet (63). In this study, the Ly6C^{hi}T-bet^{hi} effector T cells represent more terminally differentiated Th1 cells and Ly6C^{lo}T-bet^{lo} Th1 effector cells displayed a gene expression profile strikingly similar to that of memory T cells (63). Consistent with this finding, our single-cell data have revealed that Acaca was correlated well with the expression of Ly6c and Tbx21 (16).

In line with these previous observations, genetic deletion or pharmacological inhibition of ACC1 during the effector phase results in increased memory CD4$^+$ T-cell formation via alteration of the intrinsic metabolic profiles of effector T cells to resemble that of memory T cells (16). This indicates that ACC1 may control not only the Th17/Treg balance but also the CD4$^+$ T-cell effector/memory fate decision during early differentiation from naive to effector T-helper cell (Fig. 1, right).

Furthermore, segregation of CCR7 and CD137 expression during early effector CD4$^+$ T-cell differentiation influences the metabolic signature, transcriptional profile and eventual cell fate decision. Notably, most naive CD4$^+$ T cells differentiate into an ACC1^{hi}CCR7^{lo}CD137^{hi} effector cell population with a phenotype characterized by high FA biosynthesis and low mitochondrial respiratory capacity (16). This population shows a Blimp1^{hi}TCF1^{lo} expression pattern and is relatively short-lived, presumably due to failure to metabolically transition from an activated to a quiescent state during the contraction phase. Conversely, the smaller ACC1^{lo}CCR7^{hi}CD137^{lo} effector population shows a Blimp1^{lo}TCF1^{hi} profile with a phenotype characterized by low FA biosynthesis and high respiratory capacity, and this population is capable of long-term survival *in vivo* and can efficiently form memory cells.

Thus, there is increasing evidence that both CD4$^+$ and CD8$^+$ T cells with high levels of metabolic activity are characterized by reduced survival and longevity, whereas metabolic activity moderated by the blockade of mTORC1, glycolysis or *de novo* FA biosynthesis during the early effector phase can promote longevity and memory cell generation (Fig. 1, right).

Role of FA metabolism in anti-viral responses

We are just beginning to understand the role of metabolism in shaping immune function, and to reveal how modulating cell metabolism can affect immune cell differentiation. Recently, several studies have elucidated the relationship between lipid metabolism and the immune response to infectious diseases at both the cellular and organism levels (64). This topic is extremely important, because an emerging regulator of immune responses during viral infections may be a promising therapeutic target for the treatment of coronavirus disease 2019 (COVID-19).

Upon viral infection, type I interferon (IFN-α and IFN-β) is rapidly produced, which in turn induces the expression of anti-viral genes called interferon-stimulated genes (ISGs) (64). Although ISGs are critical for virus elimination, recent studies have shown that type I IFN also changes the expression of genes related to lipid metabolism, which enhances anti-viral responses (65, 66). Type I IFN signaling increases the FAO and OXPHOS required for the full activation of plasmacytoid dendritic cells (66). Furthermore, to support the host defense, the PUFA-derived lipid mediator protectin D1 attenuates influenza viral replication *via* the RNA export machinery (65). Recently, it has been shown that pharmacological inhibition of the FA biosynthesis pathway blocks severe acute respiratory syndrome coronavirus 2 (SARS-Cov-2) replication (67).

A recent study using cultured cells has shown that extrinsic supplementation with PUFAs, such as linolenic acid or eicosapentaenoic acid, inhibits both the binding of SARS-CoV-2 to human angiotensin-converting enzyme 2 receptor and viral entry into cells (68). In connection with these findings, it has been also reported that the FA profiles and activity of FA desaturase are altered in the plasma of patients with severe pneumonia induced by SARS-CoV-2 (69), suggesting that these changes promote viral replication; at the same time, however, such changes to the cell lipid metabolism are part of the usual host defense to repair cell membrane damage caused by viruses.

It has also been shown that viruses manipulate the host lipid metabolism to fuel their own replication (70, 71). For example, human cytomegalovirus alters host cell lipid metabolism to produce long-chain FAs for viral replication (72), and herpes simplex virus type 2 (a DNA virus) induces nitro-FA production, resulting in inhibition of type I IFN generation (73).

Our recent study has shown that stearoyl-CoA desaturase 2 (SCD2), which generates MUFAs, is crucial for the regulation of T-cell-mediated anti-viral responses (Fig. 2) (74). Genetic deletion of *Scd2* in CD4$^+$ T cells resulted in the reduction of the amount of MUFA-containing lipid and improved the host survival rate after lethal influenza virus infection in mice. York *et al.* have reported that suppression of the cholesterol biosynthesis pathway triggers spontaneous production of type I IFN in macrophages (75). They found that alteration of the balance between cholesterol biosynthesis and scavenging for cholesterol, rather than a reduction in the size of the endogenous lipid pool, was essential to achieve an anti-viral

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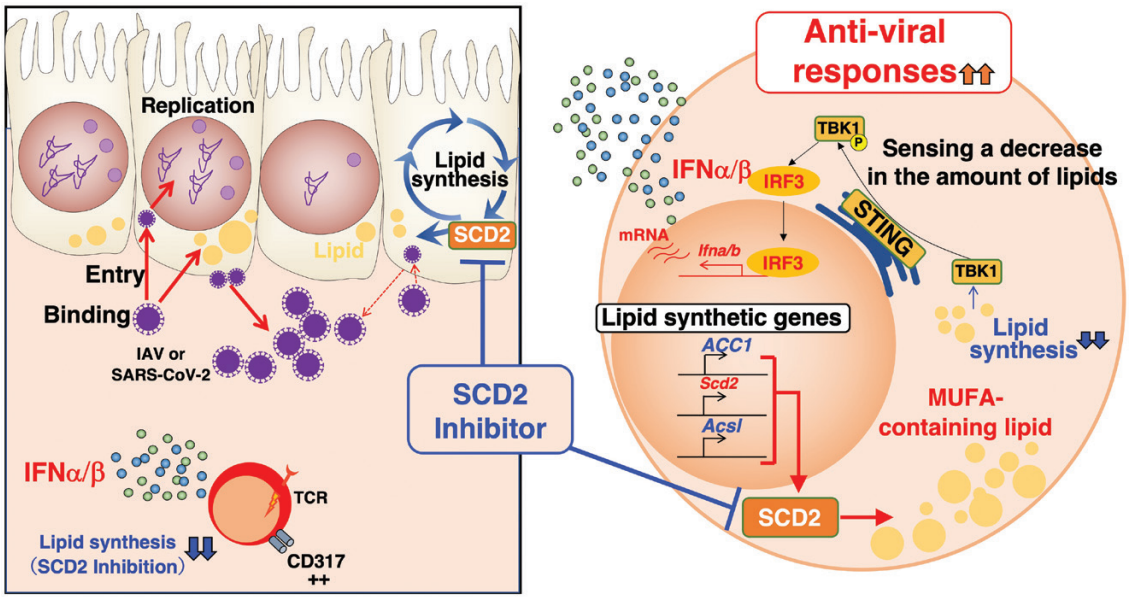

Fig. 2. Role of FA metabolism in anti-viral responses. (Left) Schematic representation depicting the relationship between lipid metabolism and viral infection. The virus life cycle is highly dependent on host cell metabolism. Viruses can manipulate the host lipid metabolism to facilitate their binding to cell membranes, entry into cells and replication. Pharmacological inhibition of host FA biosynthesis by targeting SCD2 can confer protection against infection by viruses such as influenza A virus (IAV) and SARS-CoV-2. The expression of CD317, which is a surface protein known to restrict virus infection, is increased on lung CD4$^+$ T cells in infected mice. The dotted lines show a magnified view of the cell to the right panel. (Right) The MUFA biosynthesis pathway controls the expression of ISGs in CD4$^+$ T cells. Pharmacological inhibition or genetic deletion of SCD2 leads to activation of the cGAS–STING pathway, resulting in the spontaneous production of type I IFN and anti-viral responses. Secreted type I IFN induces marked expression of ISGs. The up-regulation of ISGs can be canceled by the addition of MUFAs, indicating the importance of the FA biosynthesis pathway in the anti-viral response.

response. These data suggest that the anti-viral responses by each type of immune cell rely upon different lipid metabolic pathways.

Mechanistic insights into anti-viral responses reliant on the modulation of MUFA biosynthesis in T cells have revealed that suppression of MUFA biosynthesis induces accumulation of cytosolic DNA, resulting in activation of the cGAS–STING pathway [cyclic GMP–AMP synthase (cGAS) is a DNA sensor, and stimulator of interferon genes (STING) is the downstream effector] and the production of type I IFN. Consistent with this finding, several studies have reported that activation of the cGAS–STING pathway and subsequent anti-viral responses are controlled through lipid metabolism (75–77). York *et al.* found that the changes in the lipid membrane composition of the endoplasmic reticulum alter the sensitivity of STING to its ligand. In addition, the migration of STING from the endoplasmic reticulum to the Golgi body relies on palmitoylation at the Cys88/91 residue (76). In agreement with this, endogenously generated nitro-FAs inhibit STING palmitoylation, resulting in suppression of STING activity. Furthermore, genetic deletion of FA desaturase 2, a key enzyme in PUFA biosynthesis, has been shown to induce STING activation, and PUFA treatment has been shown to suppress STING activity, in mouse embryonic fibroblasts (77).

FA metabolism in the pathogenesis of type 2 inflammation

Another area of interest is how T-cell metabolism affects the pathogenesis of disease. Pathogenic CD4$^+$ T-cell populations drive chronic inflammatory disorders in response to cell-extrinsic environmental factors such as alarmins and cytokines (78, 79). Tissue-derived checkpoint cytokines, including IL-25, IL-33 and thymic stromal lymphopoietin (TSLP), secreted by injured epithelial cells and keratinocytes are also implicated in Th2-associated diseases through their amplification of Th2 cytokine-mediated responses (80) and their induction of IL-5 and IL-13 production by type 2 innate lymphoid cells (ILC2s).

ST2^{hi} IL-5-producing pathogenic Th2 (Tpath2) cells cause eosinophilic airway inflammation (2, 81–84). Recently, we found that ACC1-dependent FA biosynthesis plays a crucial role in Tpath2 cell differentiation and the pathogenesis of type 2 inflammation in the lung and skin (Fig. 3) (85). Genetic deletion or pharmacologic inhibition of ACC1 attenuated allergic responses during eosinophilic and basophilic inflammation. Mechanistically, mTORC1–ACC1 orchestrated cell metabolism, FA biosynthesis and maximal activation of glycolysis to facilitate IL-5 production by Tpath2 cells. Consistent with our findings, maximal aerobic glycolysis is required for cytokine production by activated T cells and does not affect cell proliferation or survival (86). We also found that IL-3 production was controlled by an epigenetic mechanism, as evident from the fact that supplementation of *Acaca*$^{\Delta T}$ Th2 cell cultures with acetate restored histone acetylation at the *Il3* promoter region. Similarly, acetate is known to enhance IFNγ production by CD8$^+$ T cells during prolonged glucose restriction (87, 88). Although the detailed mechanisms underlying how ACC1 controls the availability of cellular Ac-CoA for histone acetylation remain unclear, our previous data suggest that cellular Ac-CoA feeds into the cellular pool of metabolites used for the TCA cycle in *Acaca*$^{\Delta T}$ Th1 and Th2 cells (16).

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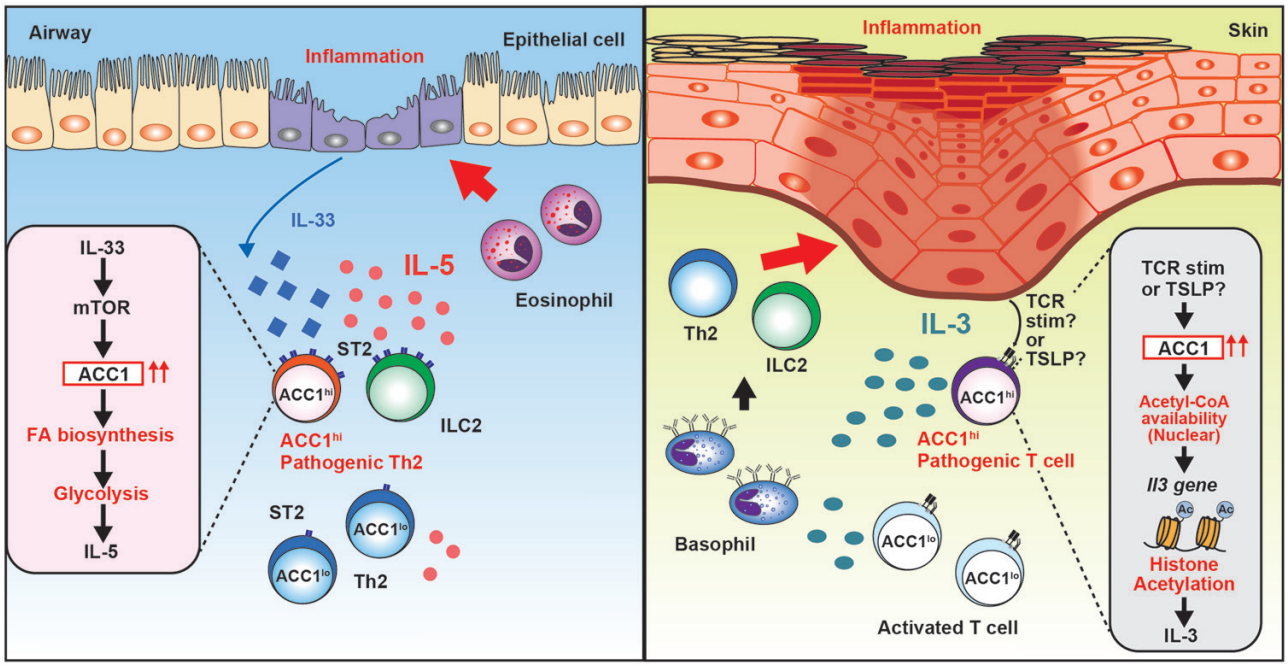

Fig. 3. Schematic diagram of airway and skin inflammation induced by ACC1^{hi} pathogenic T-helper cells. (Left) Damaged airway epithelial cells release IL-33, which induces activation of Tpath2 cells and ILC2s *via* the mTOR–ACC1 axis. ACC1-dependent *de novo* FA biosynthesis together with maximal glycolytic capacity make it possible for Tpath2 cells and ILC2s to produce a large amount of IL-5, which in turn promotes the development and activation of eosinophils to augment airway inflammation. (Right) Antigenic TCR stimulation or TSLP may activate T-helper cells to produce IL-3 *via* histone acetylation at the *Il3* gene locus. ACC1-dependent FA biosynthesis regulates the availability of cellular Ac-CoA for histone acetylation and induces IL-3 production. Furthermore, basophils activated by IL-3 prime Th2 cells and ILC2s *via* the production of IL-4 and other substances. Thereby, Th2 cells and ILC2s produce type 2 cytokines including IL-4, IL-5, IL-13 and IL-31 and induce skin inflammation.

In addition to the requirements of FA biosynthesis in Tpath2 cells, a recent study has shown that expression of the genes involved in FAO and cholesterol synthesis, including *Cpt1a*, *Adipor1*, *Aldh2*, *Cd36*, *Pdk4*, *Lpl* and *Apo* family members, is enriched within Th2 cells from the lungs and airways of a murine model of house dust mite-induced allergic airway disease (89). Furthermore, transient storage of FAs in lipid droplets has recently been shown to regulate ILC2 responses in the lung (90). That study also demonstrated esterification of FAs into triacylglycerols in ILC2s and indicated that exogenous FAs are particularly important for mediating ILC2 proliferation (90). Therefore, *de novo* FA biosynthesis, FAO and the uptake of extrinsic FAs are all required for Tpath2 cells and ILC2s to meet their lipid demands.

Conclusions

During the last few years, an interest in understanding the metabolic requirements during immune cell development has resulted in the emergence of immunometabolism as an exciting new field of research. A number of studies have demonstrated how intracellular and extracellular metabolism influence multiple aspects of immune cell development and differentiation. It is now known that FAs are essential metabolites for rapidly proliferating cells, including activated lymphocytes and tumor cells. Therefore, FA biosynthesis, catabolism and storage are integral parts of the cellular intermediary metabolism.

Recent studies have shown that many aspects of T-cell biology are deeply related to the predominant utilization of FA biosynthesis, FAO and lipid droplets. Indeed, disruption of the balance among the utilization of FA biosynthesis, FAO and lipid droplet storage in T cells has been shown to be associated with several inflammatory and anti-inflammatory conditions. Therefore, moving forward, it will be important to clarify how changes in FA metabolites affect immune cell biology during the course of inflammation and how they are influenced by different tissues and types of inflammatory signals. Furthermore, the ongoing COVID-19 pandemic has made clear the importance of elucidating the metabolic circuits that regulate anti-viral responses and T-cell biology for improving our current treatment of infectious diseases, as well as the need for improved vaccines for managing future pandemics.

In summary, a better understanding and clarification of FA metabolic regulation that dictates T-cell fate and how FA availability and microenvironmental factors influence T-cell function will provide further mechanistic insights into immune cell biology and will hopefully lead to advanced approaches for treating a variety of human disorders.

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References

1 Chang, J. T., Wherry, E. J. and Goldrath, A. W. 2014. Molecular regulation of effector and memory T cell differentiation. *Nat. Immunol.* 15:1104.
2 Nakayama, T., Hirahara, K., Onodera, A. *et al.* 2017. Th2 cells in health and disease. *Annu. Rev. Immunol.* 35:53.
3 Pulendran, B. and Ahmed, R. 2011. Immunological mechanisms of vaccination. *Nat. Immunol.* 12:509.
4 Lochner, M., Berod, L. and Sparwasser, T. 2015. Fatty acid metabolism in the regulation of T cell function. *Trends Immunol.* 36:81.
5 Corrado, M. and Pearce, E. L. 2022. Targeting memory T cell metabolism to improve immunity. *J. Clin. Invest.* 132:e148546.
6 Buck, M. D., O'Sullivan, D. and Pearce, E. L. 2015. T cell metabolism drives immunity. *J. Exp. Med.* 212:1345.
7 Bantug, G. R., Galluzzi, L., Kroemer, G. *et al.* 2018. The spectrum of T cell metabolism in health and disease. *Nat. Rev. Immunol.* 18:19.
8 Wang, R., Dillon, C. P., Shi, L. Z. *et al.* 2011. The transcription factor Myc controls metabolic reprogramming upon T lymphocyte activation. *Immunity* 35:871.
9 Jacobs, S. R., Herman, C. E., Maciver, N. J. *et al.* 2008. Glucose uptake is limiting in T cell activation and requires CD28-mediated Akt-dependent and independent pathways. *J. Immunol.* 180:4476.
10 Pollizzi, K. N. and Powell, J. D. 2014. Integrating canonical and metabolic signalling programmes in the regulation of T cell responses. *Nat. Rev. Immunol.* 14:435.
11 Angela, M., Endo, Y., Asou, H. K. *et al.* 2016. Fatty acid metabolic reprogramming via mTOR-mediated inductions of PPARgamma directs early activation of T cells. *Nat. Commun.* 7:13683.
12 Chi, H. 2012. Regulation and function of mTOR signalling in T cell fate decisions. *Nat. Rev. Immunol.* 12:325.
13 MacIver, N. J., Michalek, R. D. and Rathmell, J. C. 2013. Metabolic regulation of T lymphocytes. *Annu. Rev. Immunol.* 31:259.
14 Moremen, K. W., Tiemeyer, M. and Nairn, A. V. 2012. Vertebrate protein glycosylation: diversity, synthesis and function. *Nat. Rev. Mol. Cell Biol.* 13:448.
15 Sancak, Y., Peterson, T. R., Shaul, Y. D. *et al.* 2008. The Rag GTPases bind raptor and mediate amino acid signaling to mTORC1. *Science* 320:1496.
16 Endo, Y., Onodera, A., Obata-Ninomiya, K. *et al.* 2019. ACC1 determines memory potential of individual CD4(+) T cells by regulating *de novo* fatty acid biosynthesis. *Nat. Metab.* 1:261.
17 Endo, Y., Asou, H. K., Matsugae, N. *et al.* 2015. Obesity drives Th17 cell differentiation by inducing the lipid metabolic kinase, ACC1. *Cell Rep.* 12:1042.
18 Kidani, Y., Elsaesser, H., Hock, M. B. *et al.* 2013. Sterol regulatory element-binding proteins are essential for the metabolic programming of effector T cells and adaptive immunity. *Nat. Immunol.* 14:489.
19 Wang, R. and Green, D. R. 2012. Metabolic checkpoints in activated T cells. *Nat. Immunol.* 13:907.
20 Shi, L. Z., Wang, R., Huang, G. *et al.* 2011. HIF1alpha-dependent glycolytic pathway orchestrates a metabolic checkpoint for the differentiation of TH17 and Treg cells. *J. Exp. Med.* 208:1367.
21 Finlay, D. K., Rosenzweig, E., Sinclair, L. V. *et al.* 2012. PDK1 regulation of mTOR and hypoxia-inducible factor 1 integrate metabolism and migration of $CD8^+$ T cells. *J. Exp. Med.* 209:2441.
22 Yang, K., Shrestha, S., Zeng, H. *et al.* 2013. T cell exit from quiescence and differentiation into Th2 cells depend on Raptor-mTORC1-mediated metabolic reprogramming. *Immunity* 39:1043.
23 Duvel, K., Yecies, J. L., Menon, S. *et al.* 2010. Activation of a metabolic gene regulatory network downstream of mTOR complex 1. *Mol. Cell* 39:171.
24 Bensinger, S. J., Bradley, M. N., Joseph, S. B. *et al.* 2008. LXR signaling couples sterol metabolism to proliferation in the acquired immune response. *Cell* 134:97.
25 Forman, B. M., Tontonoz, P., Chen, J. *et al.* 1995. 15-Deoxy-delta 12, 14-prostaglandin J2 is a ligand for the adipocyte determination factor PPAR gamma. *Cell* 83:803.
26 Bettelli, E., Carrier, Y., Gao, W. *et al.* 2006. Reciprocal developmental pathways for the generation of pathogenic effector TH17 and regulatory T cells. *Nature* 441:235.
27 Zhu, J., Yamane, H. and Paul, W. E. 2010. Differentiation of effector CD4 T cell populations (*). *Annu. Rev. Immunol.* 28:445.
28 Sun, L., Fu, J. and Zhou, Y. 2017. Metabolism controls the balance of Th17/T-regulatory cells. *Front. Immunol.* 8:1632.
29 Haghikia, A., Jorg, S., Duscha, A. *et al.* 2015. Dietary fatty acids directly impact central nervous system autoimmunity via the small intestine. *Immunity* 43:817.
30 Kim, W., Barhoumi, R., McMurray, D. N. *et al.* 2014. Dietary fish oil and DHA down-regulate antigen-activated $CD4^+$ T-cells while promoting the formation of liquid-ordered mesodomains. *Br. J. Nutr.* 111:254.
31 Berod, L., Friedrich, C., Nandan, A. *et al.* 2014. *De novo* fatty acid synthesis controls the fate between regulatory T and T helper 17 cells. *Nat. Med.* 20:1327.
32 Young, K. E., Flaherty, S., Woodman, K. M. *et al.* 2017. Fatty acid synthase regulates the pathogenicity of Th17 cells. *J. Leukoc. Biol.* 102:1229.
33 Kanno, T., Nakajima, T., Kawashima, Y. *et al.* 2021. Acsbg1-dependent mitochondrial fitness is a metabolic checkpoint for tissue Treg cell homeostasis. *Cell Rep.* 37:109921.
34 Wakil, S. J., Stoops, J. K. and Joshi, V. C. 1983. Fatty acid synthesis and its regulation. *Annu. Rev. Biochem.* 52:537.
35 Abu-Elheiga, L., Matzuk, M. M., Kordari, P. *et al.* 2005. Mutant mice lacking acetyl-CoA carboxylase 1 are embryonically lethal. *Proc. Natl Acad. Sci. USA* 102:12011.
36 Endo, Y., Yokote, K. and Nakayama, T. 2017. The obesity-related pathology and Th17 cells. *Cell Mol. Life Sci.* 74:1231.
37 Gaublomme, J. T., Yosef, N., Lee, Y. *et al.* 2015. Single-cell genomics unveils critical regulators of Th17 cell pathogenicity. *Cell* 163:1400.
38 Wang, C., Yosef, N., Gaublomme, J. *et al.* 2015. CD5L/AIM regulates lipid biosynthesis and restrains Th17 cell pathogenicity. *Cell* 163:1413.
39 Santori, F. R., Huang, P., van de Pavert, S. A. *et al.* 2015. Identification of natural RORgamma ligands that regulate the development of lymphoid cells. *Cell Metab.* 21:286.
40 Hu, X., Wang, Y., Hao, L. Y. *et al.* 2015. Sterol metabolism controls T(H)17 differentiation by generating endogenous RORgamma agonists. *Nat. Chem. Biol.* 11:141.
41 Kagami, S., Owada, T., Kanari, H. *et al.* 2009. Protein geranylgeranylation regulates the balance between Th17 cells and $Foxp3^+$ regulatory T cells. *Int. Immunol.* 21:679.

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42 Field, C. S., Baixauli, F., Kyle, R. L. *et al.* 2020. Mitochondrial integrity regulated by lipid metabolism is a cell-intrinsic checkpoint for Treg suppressive function. *Cell Metab.* 31:422.
43 Weinberg, S. E., Singer, B. D., Steinert, E. M. *et al.* 2019. Mitochondrial complex III is essential for suppressive function of regulatory T cells. *Nature* 565:495.
44 Raud, B., Roy, D. G., Divakaruni, A. S. *et al.* 2018. Etomoxir actions on regulatory and memory T cells are independent of Cpt1a-mediated fatty acid oxidation. *Cell Metab.* 28:504.
45 Michalek, R. D., Gerriets, V. A., Jacobs, S. R. *et al.* 2011. Cutting edge: distinct glycolytic and lipid oxidative metabolic programs are essential for effector and regulatory CD4+ T cell subsets. *J. Immunol.* 186:3299.
46 O'Sullivan, D., van der Windt, G. J., Huang, S. C. *et al.* 2014. Memory CD8(+) T cells use cell-intrinsic lipolysis to support the metabolic programming necessary for development. *Immunity* 41:75.
47 Sukumar, M., Liu, J., Ji, Y. *et al.* 2013. Inhibiting glycolytic metabolism enhances CD8+ T cell memory and antitumor function. *J. Clin. Invest.* 123:4479.
48 Araki, K., Turner, A. P., Shaffer, V. O. *et al.* 2009. mTOR regulates memory CD8 T-cell differentiation. *Nature* 460:108.
49 Puleston, D. J., Zhang, H., Powell, T. J. *et al.* 2014. Autophagy is a critical regulator of memory CD8(+) T cell formation. *Elife* 3:e03706.
50 Xu, X., Araki, K., Li, S. *et al.* 2014. Autophagy is essential for effector CD8(+) T cell survival and memory formation. *Nat. Immunol.* 15:1152.
51 Eikawa, S., Nishida, M., Mizukami, S. *et al.* 2015. Immune-mediated antitumor effect by type 2 diabetes drug, metformin. *Proc. Natl Acad. Sci. USA* 112:1809.
52 Pearce, E. L., Walsh, M. C., Cejas, P. J. *et al.* 2009. Enhancing CD8 T-cell memory by modulating fatty acid metabolism. *Nature* 460:103.
53 van der Windt, G. J., Everts, B., Chang, C. H. *et al.* 2012. Mitochondrial respiratory capacity is a critical regulator of CD8+ T cell memory development. *Immunity* 36:68.
54 Joshi, N. S., Cui, W., Chandele, A. *et al.* 2007. Inflammation directs memory precursor and short-lived effector CD8(+) T cell fates via the graded expression of T-bet transcription factor. *Immunity* 27:281.
55 Sarkar, S., Kalia, V., Haining, W. N. *et al.* 2008. Functional and genomic profiling of effector CD8 T cell subsets with distinct memory fates. *J. Exp. Med.* 205:625.
56 Buck, M. D., O'Sullivan, D., Klein Geltink, R. I. *et al.* 2016. Mitochondrial dynamics controls T cell fate through metabolic programming. *Cell* 166:63.
57 Ross, S. H. and Cantrell, D. A. 2018. Signaling and function of interleukin-2 in T lymphocytes. *Annu. Rev. Immunol.* 36:411.
58 Pollizzi, K. N., Sun, I. H., Patel, C. H. *et al.* 2016. Asymmetric inheritance of mTORC1 kinase activity during division dictates CD8(+) T cell differentiation. *Nat. Immunol.* 17:704.
59 Chang, J. T., Palanivel, V. R., Kinjyo, I. *et al.* 2007. Asymmetric T lymphocyte division in the initiation of adaptive immune responses. *Science* 315:1687.
60 Chang, J. T., Ciocca, M. L., Kinjyo, I. *et al.* 2011. Asymmetric proteasome segregation as a mechanism for unequal partitioning of the transcription factor T-bet during T lymphocyte division. *Immunity* 34:492.
61 Verbist, K. C., Guy, C. S., Milasta, S. *et al.* 2016. Metabolic maintenance of cell asymmetry following division in activated T lymphocytes. *Nature* 532:389.
62 Chen, Y. H., Kratchmarov, R., Lin, W. W. *et al.* 2018. Asymmetric PI3K activity in lymphocytes organized by a PI3K-mediated polarity pathway. *Cell Rep.* 22:860.
63 Marshall, H. D., Chandele, A., Jung, Y. W. *et al.* 2011. Differential expression of Ly6C and T-bet distinguish effector and memory Th1 CD4(+) cell properties during viral infection. *Immunity* 35:633.
64 McNab, F., Mayer-Barber, K., Sher, A. *et al.* 2015. Type I interferons in infectious disease. *Nat. Rev. Immunol.* 15:87.
65 Morita, M., Kuba, K., Ichikawa, A. *et al.* 2013. The lipid mediator protectin D1 inhibits influenza virus replication and improves severe influenza. *Cell* 153:112.
66 Wu, D., Sanin, D. E., Everts, B. *et al.* 2016. Type 1 interferons induce changes in core metabolism that are critical for immune function. *Immunity* 44:1325.
67 Chu, J., Xing, C., Du, Y. *et al.* 2021. Pharmacological inhibition of fatty acid synthesis blocks SARS-CoV-2 replication. *Nat. Metab.* 3:1466.
68 Goc, A., Niedzwiecki, A. and Rath, M. 2021. Polyunsaturated ω-3 fatty acids inhibit ACE2-controlled SARS-CoV-2 binding and cellular entry. *Sci. Rep.* 11:5207.
69 Pérez-Torres, I., Guarner-Lans, V., Soria-Castro, E. *et al.* 2021. Alteration in the lipid profile and the desaturases activity in patients with severe pneumonia by SARS-CoV-2. *Front. Physiol.* 12:667024.
70 Heaton, N. S. and Randall, G. 2011. Multifaceted roles for lipids in viral infection. *Trends Microbiol.* 19:368.
71 Schoggins, J. W. and Randall, G. 2013. Lipids in innate antiviral defense. *Cell Host Microbe* 14:379.
72 Purdy, J. G., Shenk, T. and Rabinowitz, J. D. 2015. Fatty acid elongase 7 catalyzes lipidome remodeling essential for human cytomegalovirus replication. *Cell Rep.* 10:1375.
73 Hansen, A. L., Buchan, G. J., Rühl, M. *et al.* 2018. Nitro-fatty acids are formed in response to virus infection and are potent inhibitors of STING palmitoylation and signaling. *Proc. Natl Acad. Sci. USA* 115:E7768.
74 Kanno, T., Nakajima, T., Yokoyama, S. *et al.* 2021. SCD2-mediated monounsaturated fatty acid metabolism regulates cGAS-STING-dependent type I IFN responses in CD4(+) T cells. *Commun. Biol.* 4:820.
75 York, A. G., Williams, K. J., Argus, J. P. *et al.* 2015. Limiting cholesterol biosynthetic flux spontaneously engages type I IFN signaling. *Cell* 163:1716.
76 Mukai, K., Konno, H., Akiba, T. *et al.* 2016. Activation of STING requires palmitoylation at the Golgi. *Nat. Commun.* 7:11932.
77 Vila, I. K., Chamma, H., Steer, A. *et al.* 2022. STING orchestrates the crosstalk between polyunsaturated fatty acid metabolism and inflammatory responses. *Cell Metab.* 34:125.
78 Saravia, J., Chapman, N. M. and Chi, H. 2019. Helper T cell differentiation. *Cell Mol. Immunol.* 16:634.
79 Walker, J. A. and McKenzie, A. N. J. 2018. T(H)2 cell development and function. *Nat. Rev. Immunol.* 18:121.
80 Holgate, S. T. 2012. Innate and adaptive immune responses in asthma. *Nat. Med.* 18:673.
81 Endo, Y., Hirahara, K., Iinuma, T. *et al.* 2015. The interleukin-33-p38 kinase axis confers memory T helper 2 cell pathogenicity in the airway. *Immunity* 42:294.
82 Endo, Y., Hirahara, K., Yagi, R. *et al.* 2014. Pathogenic memory type Th2 cells in allergic inflammation. *Trends Immunol.* 35:69.
83 Yamamoto, T., Endo, Y., Onodera, A. *et al.* 2018. DUSP10 constrains innate IL-33-mediated cytokine production in ST2(hi) memory-type pathogenic Th2 cells. *Nat. Commun.* 9:4231.
84 Obata-Ninomiya, K., Ishiwata, K., Nakano, H. *et al.* 2018. CXCR6(+)ST2(+) memory T helper 2 cells induced the expression of major basic protein in eosinophils to reduce the fecundity of helminth. *Proc. Natl Acad. Sci. USA* 115:E9849.
85 Nakajima, T., Kanno, T., Yokoyama, S. *et al.* 2021. ACC1-expressing pathogenic T helper 2 cell populations facilitate lung and skin inflammation in mice. *J. Exp. Med.* 218:e20210639.
86 Chang, C. H., Curtis, J. D., Maggi, L. B. *et al.* 2013. Posttranscriptional control of T cell effector function by aerobic glycolysis. *Cell* 153:1239.
87 Qiu, J., Villa, M., Sanin, D. E. *et al.* 2019. Acetate promotes T cell effector function during glucose restriction. *Cell Rep.* 27:2063.
88 Vodnala, S. K., Eil, R., Kishton, R. J. *et al.* 2019. T cell stemness and dysfunction in tumors are triggered by a common mechanism. *Science* 363:eaau0135.
89 Tibbitt, C. A., Stark, J. M., Martens, L. *et al.* 2019. Single-cell RNA sequencing of the T helper cell response to house dust mites defines a distinct gene expression signature in airway Th2 cells. *Immunity* 51:169.
90 Karagiannis, F., Masouleh, S. K., Wunderling, K. *et al.* 2020. Lipid-droplet formation drives pathogenic group 2 innate lymphoid cells in airway inflammation. *Immunity* 52:620.

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